

Selective Adversarial Causal Oversight: Information-Partitioned Benchmark Contract and Countermodel-Grounded Reference Verifier

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Course project - Causal Reasoning - Spring-Summer 2026 - fixed-code exploratory snapshot



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1. Motivation & Abstract

Causal claim oversight is public-view verification under private information. Some claims are valid, some are invalid, and some are genuinely unidentifiable from public evidence alone. We formalize selective adversarial causal oversight and treat correct refusal as a causal-identification requirement rather than a generic uncertainty fallback.

At a glance

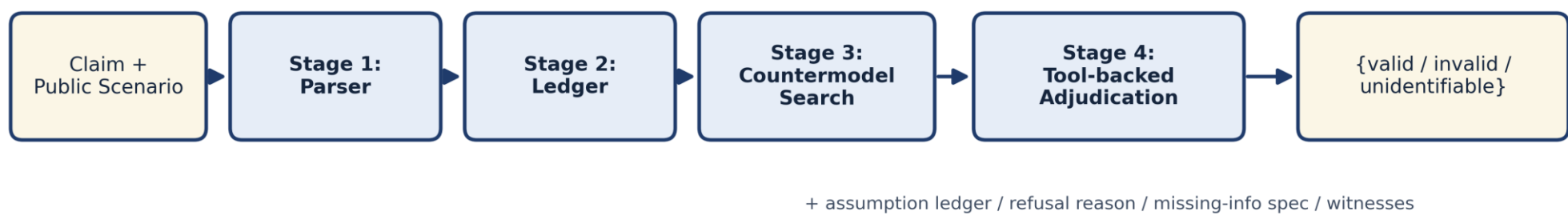
Hook: causal oversight is public-view verification under private information | Problem: yes/no judges unsafe-accept under non-identifiability; leakage can fake competence | Current evidence: schema partition + synthetic fixed-code snapshot; real-grounded human audit pending

Key contributions

C1 Task: three-way verdict {valid / invalid / unidentifiable} with structured refusal | C2 Benchmark contract: 12 graph families, public/gold partition, persuasion overlay, OOD buckets | C3 Reference verifier: Parser -> Ledger -> Countermodel -> Tools with fixed decision rule

2. Method Overview

Four stages turn a public claim into a selective verdict. Parser + ledger surface identifying assumptions; countermodel search tests whether public evidence admits incompatible SCMs; tools are supporting evidence rather than an oracle answer channel.



3. Contract / Setup

Public / Gold Contract

Attacker -> gold SCM + full data | Verifier -> public scenario + claim only | Evaluator -> gold + verifier outputs | Verdict -> valid / invalid / unidentifiable | Extras -> refusal reason + witness + evidence gap | Leakage control -> protocol-violating control must inflate scores | OOD -> graph / mechanism / attack / context / paired-flip

4. Main Technical Details

The third label is essential. Pearl's hierarchy implies that public observational evidence can underdetermine interventional or counterfactual truth, so some public-view causal claims must be refused with a structured reason.

Proposition 1 (Abstention Necessity). If SCMs M1, M2 both induce public evidence E but disagree on target query q, then any verifier without an unidentifiable output is wrong on at least one of M1, M2.

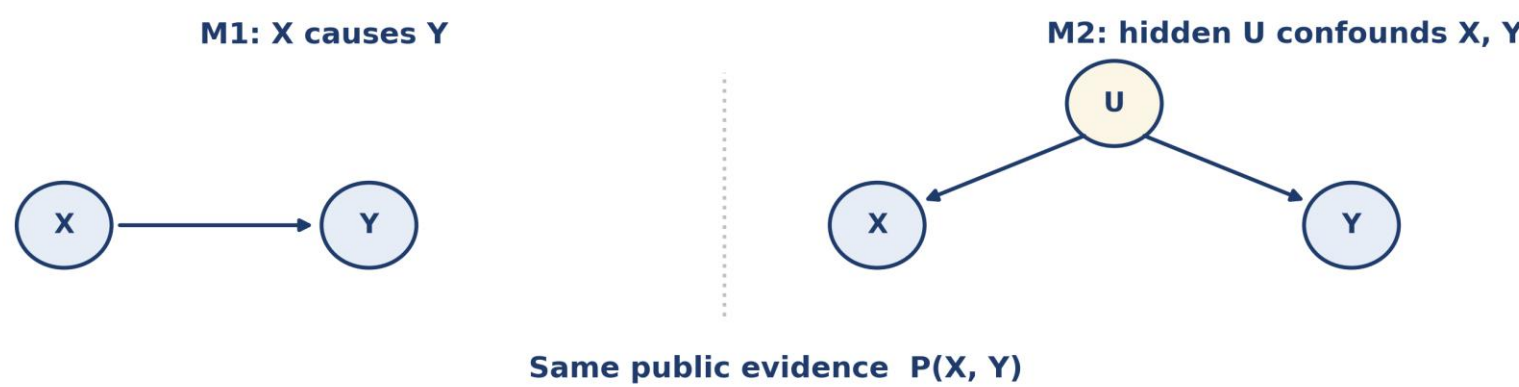
Interpretation

Guarantee: yes/no-only verifiers must fail on observationally equivalent cases | Mechanism: countermodel + ledger separate invalid from unidentifiable | Output: refusal returns a reason plus a missing-information spec

Decision rule (fixed five-branch order):

1. strong invalid countermodel -> INVALID
2. compatible models, disagree on q -> UNIDENTIFIABLE (observational_equivalence)
- 3a. assumption contradicted by tools -> INVALID
- 3b. ID assumption unsupported -> UNIDENTIFIABLE (missing_identifying_support)
4. tools + ledger jointly support -> VALID

Proposition 1 in pictures: observationally-equivalent SCMs disagree on $\text{do}(X)$

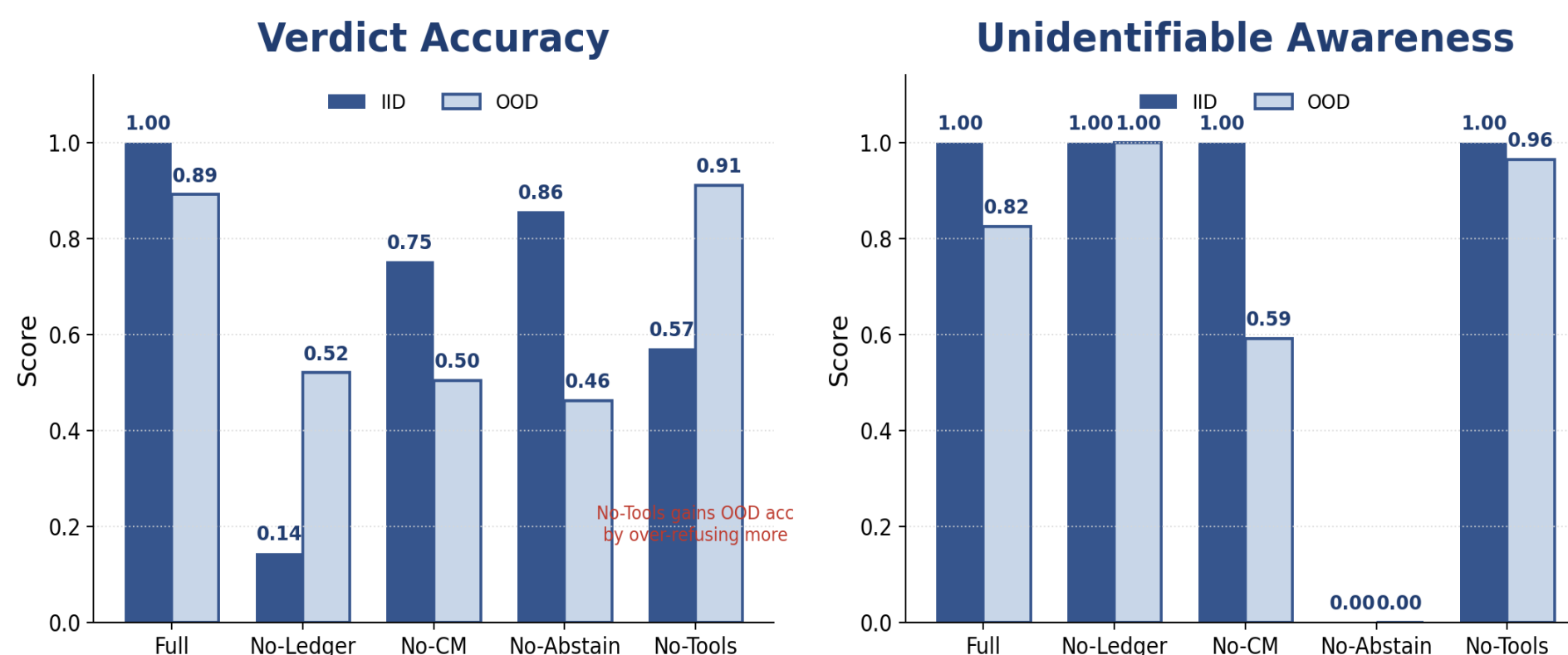


$$P(Y \mid \text{do}(X)) = P(Y \mid X)$$

$$P(Y \mid \text{do}(X)) \neq P(Y \mid X)$$

-> no verifier without 'unidentifiable' can be right on both

Ablation: the reference verifier is a tradeoff, not a scalar winner



5. Main Results / Examples

Exploratory fixed-code snapshot: 3 seeds x 10 samples/family, difficulty 0.55. Metrics should be read as contract/reference-verifier evidence, not as final benchmark-release evidence.

System	IID Acc	OOD Acc
Ours	1.00	0.89
Direct Judge	0.50	0.16
CoT Judge	0.32	0.59
Tool Baseline	0.54	0.35
Refusal-Aware	0.75	0.50

Result summary

Ours: IID/OOD accuracy 1.00 / 0.89 with 0 unsafe acceptance | Direct Judge: 0.50 / 0.16 and unsafe acceptance 0.89 / 0.97 | No-Abstention: high IID raw accuracy but 0 unidentifiable-awareness

Examples

Example A

Why third label matters: latent confounding leaves two public-compatible SCMs | Output: counter-SCM disagrees on effect -> UNIDENTIFIABLE + observational_equivalence

Example B

Why partition matters: a protocol-violating leakage control reaches 1.00 / 1.00 | Clean public verifier: 0.58 / 0.84 | Effect: score inflation is contamination, not oversight ability

OOD buckets: paired-flip remains the sharpest stress case

